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Age, palaeomagnetism and chemistry of melilite basalts in the Southern Cape, South Africa

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Summary. Post-Karroo melilite basalts intrude Cretaceous and older sedimentary units in the Cape Province, South Africa. K–Ar ages determined from 4 occurrences range from 63–75 Ma b.p. (Late Cretaceous to earliest Tertiary) and are significantly older than alkaline intrusions to the S, on the Alphard Bank. These igneous intrusions may be part of a linear trend that connects young oceanic volcanism at Bouvet Island with Cretaceous kimberlitic activity in southern Africa. This lineament nearly parallels the Walvis Ridge and a hot spot model for its origin is examined.

1. Introduction

The occurrence at several localities in the southern and western Cape Province, South Africa, of Post-Karroo intrusions of melilite basalt and related ultramafic rocks, has been known for many years (Taljaard, 1937; Du Toit, 1954) (Fig. 1). Recently, undersaturated alkalic igneous rocks have been discovered offshore on the Alphard Bank (Dingle & Gentle, 1972). Reflection seismic surveys have identified numerous small intrusions which, from dredged samples, comprise olivine basalts, trachybasalts and trachytes. Dingle & Gentle (1972) reported a K-Ar age of 58±2.4 Ma from sanidine separated from a sample of aegirine-augite trachyte. Because of the chemical similarity, geographical proximity and presumed contemporaneity of the offshore and onshore intrusions, Dingle and Gentle proposed the term Alphard Tertiary Igeneous Province to describe this igneous event.

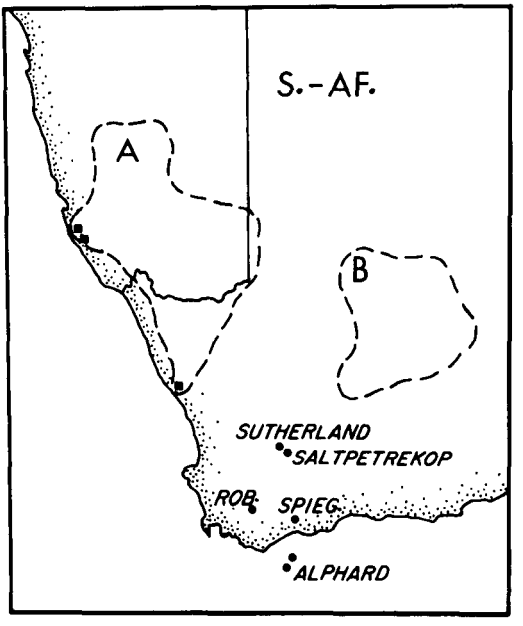


Figure 1. Late Cretaceous to early Tertiary melilite basalts crop out in the Cape Province, South Africa, at Sutherland, Saltpetrekop, Robertson and Spiegel River. Similar alkalic volcanism has been recognized to the S, offshore on the Alphard Bank. Diamond-bearing kimberlitic pipes of Cretaceous age exist in a diffuse pattern to the northeast (B) and scattered Tertiary alkalic volcanism covers a large region to the northwest (A).

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Available geochronology for phonolites and olivine melilitites outcropping in the western Cape Province, however, suggests a significantly younger phase of igneous activity. Kroner (1973) reported an age of 38.5 Ma for an intrusive melilitite which, from the location he provided (30° 43′ S, 17° 59′ E), may be related to the Bushmanland occurrences reported by Taljaard (1937). Similar ages (35.7 and 37 Ma) obtain from phonolitic and melilititic plugs in the Klinghardt Mountains in southwest Africa, some 400 km to the NW (Kroner, 1973; pers. comm., J. A. S. Adams).

In view of this age discrepancy, it is not clear whether the alkalic intrusions of the western Cape Province are genetically related to those of the Alphard Bank. This paper reports radiometric ages, new chemical analyses, and paleomagnetic field directions from 4 of the onshore intrusions in the southern Cape Province. The possible relationship of this igneous activity to Tertiary volcanism in southern Africa is examined.

2. Geology and tectonic setting

Taljaard (1937) gave a comprehensive account of the geology and petrography of the melilite basalts of the southern Cape, and these data are incorporated in a regional summary by Du Toit (1954).

The principal occurrences, and the 4 localities sampled in this study, are shown in Figure 1. The basalts sampled at Spiegel River (or Heidelberg) and Robertson (localities 3 and 1) appear to be from relatively isolated, small (approximately 200 m diameter) intrusive plugs. On Sutherland Commage (locality 4) the field relationships are more obscure and it is uncertain whether the outcrops represent sheet-like bodies or a single plug or ring dike (Taljaard, 1937, p. 291). Saltpetrekop (locality 2) is a conspicuous peak about 15 km SE of Sutherland, which is formed by a plug of breccia and tuff. Numerous satellitic pipes and dikes are grouped around this central volcano. Here, trachyte is the most common rock type, but the sequence of intrusion is: olivine melilitite, trachyte, and carbonatite (McIver & Ferguson, in prep., after de Wet, 1974, unpubl., thesis). One of the satellite pipes, at Matjesfontein west of Saltpetrekop, contains large aggregates or crystals of hornblende, ilmenite and biotite (Du Toit, 1954, p. 425; McIver & Ferguson, in prep.). No outcrop remains at Matjesfontein, but in the waste removed from a now-abandoned shaft, fragments of these materials can be found. An age is reported from a 4 cm biotite-mica book collected at this locality. About 5 km N of Saltpetrekop, a small (100 m diameter) melilitite plug crops out and was sampled for paleomagnetic study.

3. Petrography

The olivine melilitites from Sutherland Commage and Saltpetrekop are distinct in thin section from those from Robertson and Spiegel River. Olivine (up to 30%, Taljaard, 1937) in the melilite basalts from Sutherland and Saltpetrekop are up to 1 cm in hand specimen and shiny black, sometimes with a green core. Most crystals are euhedral to subhedral and show an even extinction. Rare xenomorphic olivines (and aggregates) show blocky and undulating extinction and are interpreted as xenocrysts from disintegrated ultramafic nodules. A similar interpretation applies to rare red spinel with titanomagnetite outgrowth. The forsterite content of euhedral olivines ranges from 82 to 89.5. The original euhedral shape of the olivines is commonly outlined by small crystals of perovskite and titanomagnetite.

Olivine reacted with the melt at the time or shortly before the groundmass crystallized and monticellite and biotite were formed, both of which grow poikilitically into the groundmass. Before this reaction, olivine crystallized together with large laths of melilite (olivine does not exhibit the groundmass reaction to monticellite where it is in direct contact with melilite). Mg

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Table 1. Composition of South African Olivine melilitites (all normalized to 100%)

	SALT 1	SALT 6	SUT-2	SUT-7	SPIEG-1	SPIEG-7	A	ROB-2	ROB-6	B
SiO_2	36.2	37.2	33.3	32.1	38.0	38.4	36.2	37.6	38.3	37.1
TiO_2	2.4	2.2	3.2	2.9	4.3	4.2	2.3	4.1	4.6	3.6
Al_2O_3	7.2	7.8	7.3	6.9	7.4	7.5	15.2	7.4	7.4	10.6
Fe_2O_3	3.2	3.8	6.0	5.9	6.3	5.3	13.9	5.5	7.1	12.4
FeO	8.0	6.3	7.3	8.4	7.8	8.5	—	9.7	9.1	—
MgO	19.5	19.6	19.6	20.3	17.3	17.6	13.6	18.5	16.8	19.1
CaO	18.2	17.3	16.7	18.4	12.9	12.4	11.4	12.5	12.7	13.0
Na_2O	2.4	2.3	2.6	2.1	2.8	2.6	2.4	2.0	1.3	1.3
K_2O	1.0	1.5	2.0	0.9	2.1	2.2	1.8	1.7	1.3	0.5
P_2O_5	1.6	1.9	1.9	2.0	0.9	1.0	—	0.9	1.2	
L.O.I.	3.7	4.7	3.7	2.8	1.2	1.3	—	1.0	1.9	

TPD microprobe analyses from glasses made with an iridium strip-heater, average of 6–8 analyses, beam ~80 μm. Analyses A and B from Taljaard (1937).

values ($= 100 \times Mg/(Mg+Fe^{+2})$) of olivine and melilite are similar (87–89) and two samples from Saltpetrekop clearly show cumulate olivine and melilite.

Melilite continued to crystallize during formation of the groundmass together with titanomagnetite, perovskite, nepheline and sodalite. The rest of the groundmass is comprised of zeolite, calcite and strongly pleochroic (green to light brown) biotite, which also form rims around larger, brownish biotite flakes. Analcime was not found and Taljaard (1937) may have mistaken it for sodalite. Low totals and a strong Cl-peak in the X-ray spectrum of a TPD electron microprobe suggest 5–6% Cl in sodalite. McIver & Ferguson (in prep.) describe anorthite in the groundmass of samples from Saltpetrekop and we have found minute wollastonite needles and very rare, small, Si-rich, and Ti-Al-poor clinopyroxenes.

In the Robertson and Spiegel River samples, euhedral olivine phenocrysts (forsterite 82–85.7 in ROB 2, 81.6–87.1 in ROB 6, 84.3–90.9 in SPIEG 1, and 83.3–87.7 in SPIEG 7) and microphenocrysts of melilite (Mg values 75–77), perovskite and titanomagnetite are set in a groundmass of clinopyroxene, nepheline, titanomagnetite and rare biotite. Rare xenomorphic olivines with undulating extinction and red brown spinel also occur and are interpreted as fragments from ultramafic nodules.

4. Chemistry

Chemical analyses of 2 separate samples from each of the four melilite basalt intrusions sampled for paleomagnetics have been obtained (Table 1). The powdered samples were melted with an iridium strip-heater and quenched to a clear, brown glass (the Robertson and Spiegel River glasses contained rare quench rutile). Duplicates from each sample were made and analysed for major elements (except P_2O_5) with a TPD microprobe (Reed & Ware, 1973, 1975). The beam was defocused to about 80 μm and 6–8 analyses were made from each glass. P_2O_5 was determined on pressed powder pellets by XRF, while FeO was determined by spectrophotometry (Kiss, 1974). The analysis, including P_2O_5 and FeO, Fe_2O_3 are normalized to 100% in Table 1. Loss on ignition was measured by heating the powdered samples to 1000 °C for 2 h.

As in thin sections, clear chemical differences can be seen between the Sutherland, Saltpetrekop melilitites on one hand, and the Robertson, Spiegel River rocks on the other. The Sutherland Commonage melilite basalts show very low Si-contents and high Mg and Ca contents (Table 1). They are similar to analyses 6, 7 and 9 of McIver & Ferguson (in prep.). These authors also report more silicic compositions (analyses 5 and 8), similar to the analyses from Saltpetrekop in Table 1. The Saltpetrekop rocks have similar MgO and CaO, but lower TiO_2 contents and

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higher Mg values than the melilitites from Sutherland. They can be interpreted as cumulates (of olivine and melilite – see above) of a parental magma, similar in composition to the rocks from Sutherland Commonage given in Table 1.

The Spiegel River and Robertson olivine melilitites are SiO_2-and especially TiO_2-richer and MgO, P_2O_5 and especially CaO-poorer than the Sutherland and Saltpetrekop rocks. The differences between these new analyses and those of Taljaard (1937, table 1) are not understood; possibly the contrast in Al_2O_3 content is related to the difficulty of Al_2O_3 determination by wet chemical analysis.

Fragments of ultramafic nodules and high Mg values make the olivine melilitites from the Cape Province likely candidates for primary magmas. Experiments on an olivine melilitite similar in composition to those from Saltpetrekop and Sutherland show that those rocks may be derived by partial melting of a garnet periodotite with small amounts of H_2O and CO_2 at about 35 kb and 1130 °C (Brey, 1977). Less undersaturated olivine melilitites like those from Spiegel River and Robertson originate at lower pressures (but not less than 26 kb), higher degrees of partial melting and greater H_2O/CO_2 in the source region (Brey & Green, 1977; Brey, 1977).

5. Paleomagnetic sampling and measurement

From 6 to 8 hand samples, collected at each of the 4 localities, were orientated either with a sun compass (Creer & Sanver, 1967) or by reference to topographic prominences (Fig. 1, Table 2). Core samples, 1 in. in diameter were drilled on a drill press, and orientated prior to removal. At least one specimen from each sample was demagnetized in alternating fields (AF) in seven or 8 steps to 500 or 700 oe. Based on these results, a second specimen from each sample was demagnetized at 150 and 300, and in some cases 500–700 oe.

The results, summarized in Table 2, are however extremely disappointing. Only the samples from the Robertson body gave reasonably consistent, stable results indicating magnetization in a reversed field. The Saltpetrekop samples were of mixed polarity, 4 reversed and 2 apparently normal. The mean after arbitrarily reversing the direction of the 2 normal samples, is marginally significant.

The Spiegel River samples, after intense AF demagnetization, clustered into 2 groups, with rather peculiar mean directions of dubious significance. The Sutherland Commonage samples were all hopelessly scattered after demagnetization.

As the Robertson body is relatively small; and presumably cooled quite quickly, there is no reason to assume that secular variation has been averaged in the vector mean obtained. So this result has little paleomagnetic value, other than to indicate that the body was emplaced at a time when the earth's field was reversed. The apparently mixed polarities recorded at Saltpetrekop and Spiegel River are simply not understood.

It is noteworthy (Table 2) that the samples from the Robertson body have a Median Demagnetizing Field (M.D.F. – alternating field which reduced NRM to 50% of its original intensity), much higher (~80 oe) than all of the others (~40 oe); this evidence of higher stability of NRM is in keeping with the more consistent results obtained from this intrusive body.

6. Geochronology

Whole rock samples for conventional K–Ar age determinations were selected on the basis of thin-section examination. Generally, these samples are holocrystalline, medium-grained basalts, devoid of glass, which exhibit only minor secondary clay mineralization. Three whole rock samples, one each from the Sutherland Commonage, Spiegel River and Robertson localities, appear sufficiently fresh to attempt age determinations. The biotite book from Matjesfontein,

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Table 2. Paleomagnetic data for melilite basalts from the Cape Province, South Africa

Site	Name	Coordinates	N	D	I	k	J_n	X	M.D.F.	O.D.F.	N	D	I	k	$\alpha 95$
1	Robertson	33.9 S, 20.0 E	6	198.1	74.9	18.2	11.5^{-3}	4.4^{-3}	80 oe	150	6	180.4	66.3	28.7	12.7
2	Saltpetrekop	32.4 S, 20.9 E	6	181.7	−62.3	1.9	8.7^{-3}	3.7^{-3}	30	150	6[1]	173.3	45.0	7.5	26.1
3	Spiegel River	34.0 S, 21.1 E	7	20.9	21.1	2.6	22.6^{-3}	6.1^{-3}	40	700	4[2]	26.6	7.4	27.3	17.9
4	Sutherland	32.4 S, 20.6 E	6	161.2	37.3	1.7	15.3^{-3}	5.2^{-3}	40[3]	700	3[2]	142.0	82.6	19.4	28.8

1. Direction of 2 samples (1 and 4) have been arbitrarily reversed
2. Samples from site 3 separated into two distinct groups: 1, 2, 3, 5, and 4, 6, 7.
3. All demagnetized results widely scattered.

N = number of samples.
D = declination.
I = inclination.
k = Fisher's dispersion parameter.
$\alpha 95$ = cone of 95% confidence.
J_n = intensity of NRM emu/cc.
X = magnetic susceptibility emu/cc. oe.
M.D.F. = median demagnetizing field – AC field required to reduce J_n by 50%.
O.D.F. = optimal demagnetizing field – that demagnetizing field after which the best statistical grouping was obtained.

in the Saltpetrekop area was also judged as unlikely to have suffered argon loss. Table 3 summarizes the geochronology for these four occurrences.

Whole rock samples were crushed and the 1.5–3 mm size fraction was split into 2 aliquants for potassium and argon analysis. Potassium was determined in duplicate by flame photometry. Argon was extracted by fusing the crushed sample in a high-vacuum glass line equipped with molecular sieve and gettering furnaces (McDougall, 1966). Measurement of the isotopic composition of argon was made in the static mode using a modified commercial gas source mass spectrometer (AEI MS-10) with 4.2 kG permanent magnet and digital readout, operated at the Australian National University. Isotope dilution, adding a known amount of ^{38}Ar during sample fusion, allowed calculation of the absolute amount of radiogenic argon. Data reduction, age calculation and error analysis were done by computer following the method described by McDougall, Polach & Stipp (1969). Quoted uncertainties are ±2 standard deviations.

Age determinations fall into two groups. Ages of melilite basalt samples from Spiegel River (62.6±1.0 Ma) and Robertson (63.7±1.3 Ma) are indistinguishable, within limits of detection. A similarly consistent group is formed by ages of the Matjesfontein (Saltpetrekop) mica (74.7±3.0 Ma) and Sutherland (74.9±1.3 Ma). All samples contain considerable potassium and provide a uniformly high percentage of radiogenic argon. The consistency of ages within both groups over a range of potassium contents suggests that the younger ages at Spiegel River and Robertson are probably real and not the result of argon loss. Because of the close agreement of the biotite ages from Matjesfontein and the whole rock ages from Sutherland for very different potassium contents, excess argon cannot have made a significant contribution to the older ages of those intrusions. We conclude that the range in ages exhibited by these 4 intrusive plugs or pipes is real and manifests intermittent alkalic volcanism between 63 and 75 Ma ago in the southern Cape Province.

A single earlier geochronological study of one of these intrusions is due to Holmes & Paneth (1936). They report an age of 51 Ma for the melilite basalt at Spiegel River determined by the helium–uranium–thorium method. They believe this age to underestimate the true age of the intrusion by 10–25%, bringing it closer to the ages reported here.

7. Discussion

It has been proposed (Carey, 1958; Wilson, 1963; Morgan, 1972) that aseismic ridges, stretching from continental margins to mid-ocean spreading ridges, are volcanic manifestations of sublithospheric hot spots (plumes) which thermally and mechanically pierce the overriding crust. The Walvis Ridge and the Iceland–Faeroes Ridges are suggested examples of this activitiy which mark the path of lithospheric plates over supposedly fixed volcanic sources in the Earth's mantle.

Bathymetric data (Johnson, Hey & Lowrie, 1973) reveal elements of a linear feature which extends SW from the Tertiary intrusions of the Cape Province and the Alphard Bank to Bouvet Island on the Atlantic–Indian Rise (Fig. 2). Bouvet Island is a very young volcanic feature (Verwoerd, 1972) proposed by Morgan (1972) to be the present site of hot spot activity. If this is the case, the aseismic ridge (in places only intermittent seamounts) connecting Bouvet to the Alphard Bank could have resulted from northeastward movement of the African plate, away from the Bouvet plume, during the last 60 Ma. Ages reported here suggest that the Cape Province melilite basalts may be even earlier manifestations of the Bouvet plume, intruded 63–75 Ma ago. The conspicuous Gough–Walvis Ridge lineament, elsewhere proposed to be a manifestation of hot spot activity, very nearly parallels the Bouvet–Alphard Province trend. Such a geometry would be expected from rigid plate motion over stationary sources of volcanism. It is noted also that these lineaments are oblique to, rather than parallel to, the recognized pattern of transform faulting in the SE Atlantic, implying that this anomalous volcanism cannot be attributed to 'leaky' transform faults.

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Table 3. Geochronology of melilite basalts from the Cape Province, South Africa

Number	Location and description	%K	$100\times{}^{40}Ar_{rad}$		
			$^{40}Ar_{rad}$ ($\times10^{-10}$ moles/g)	$^{40}Ar_{total}$	Age (M.Y.) ±S.D.
74–379	Spiegel River (34.0° S, 21.1° E) melilite basalt	0.900; 0.896	1.0174	79.6	62.6±1.0
74–380	Robertson (33.9° S, 20.0° E) melilite basalt	1.490; 1.489	1.714	37.0	63.7±1.3
74–382 A	Saltpetrekop (32.4° S, 20.8° E) biotite	6.679; 6.650	9.2995	81.3	76.8±1.3
			8.7685	75.6	72.5±1.2
74–385	Sutherland (32.4° S, 20.6° E) melilite basalt	1.656; 1.649	2.2187	95.0	73.9±1.3
			2.2748	93.7	75.8±1.2

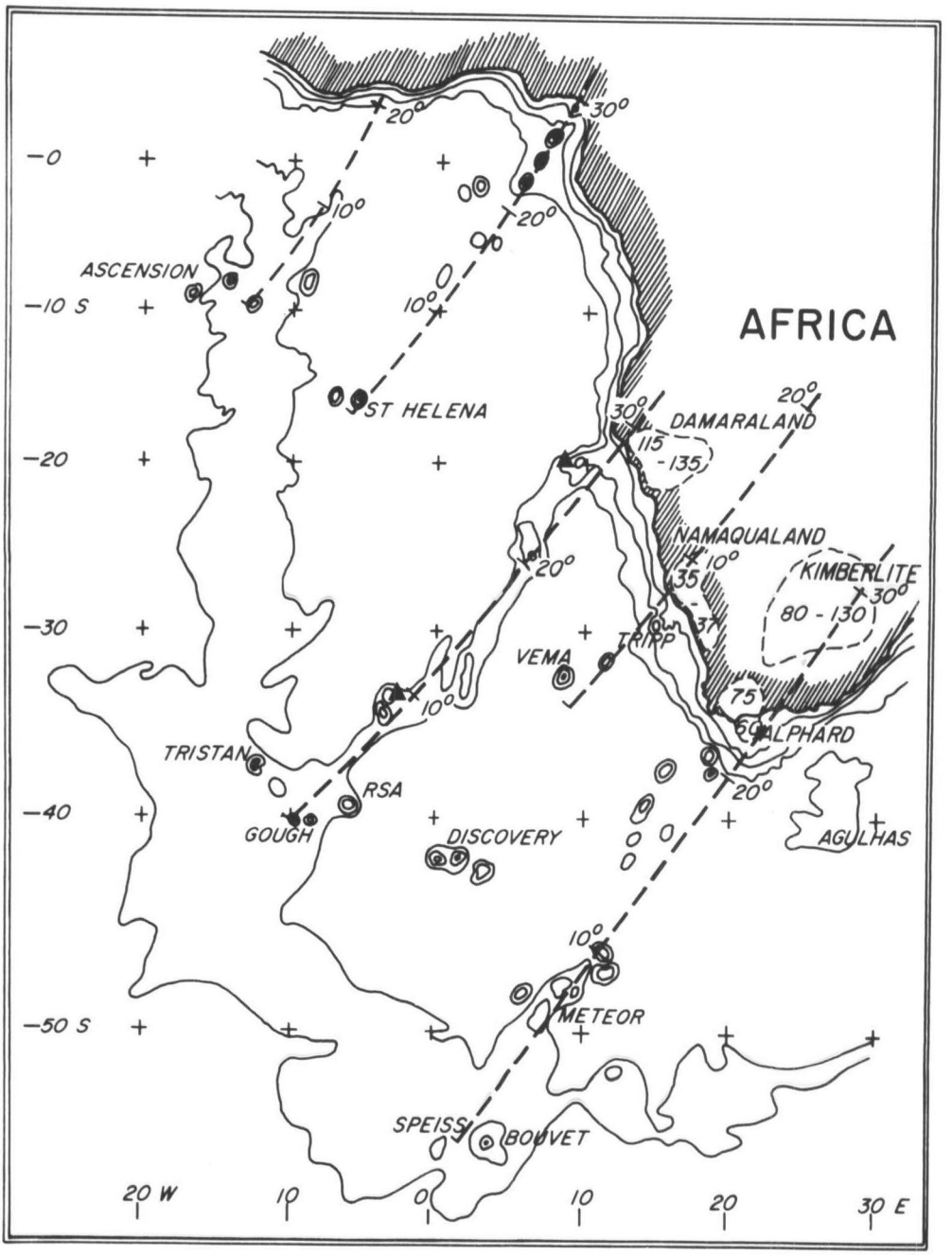


Figure 2. Bathymetry for the southeast Atlantic Ocean is illustrated by 4, 2 and 1 km depth contours. Cretaceous and Tertiary alkalic volcanism in southern Africa may be related to subparallel lineaments of oceanic islands and seamounts which result from hotspot volcanism. Small circles of latitude, constructed about a pole of rotation at 22° 30′ N, 47° 30′ W, are superimposed on the lineaments. Ten-degree intervals of rotation are marked. Solid triangles are DSDP drilling sites, of basement age 38 and 105 Ma, respectively (Scientific Staff, DSDP, 1975), while ages for continental volcanism are given in millions of years and are from Siedner & Miller (1968), Kroner (1973), Allsop & Barrett (1975), Dingle & Gentle (1972) and this study. If this cratonic igneous activity is related to the passage of the African plate over mantle hotspots, its manifestation on continental crust is much more varied and complex than on oceanic crust.

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The linear trend of volcanism from Bouvet to the Cape Province basalts may continue further inland to the NE, possibly indicated by the numerous kimberlite pipes, concentrated near Kimberley, and in Lesotho (Moore, 1975). Many of these, according to Allsopp & Barrett (1975), were emplaced about 84–90 Ma ago.

Melilite basalts and kimberlites have common petrochemical attributes, and a genetic relationship between them has been postulated (see Yoder, 1975; Brey, 1977). An extrapolation from the Cape melilite basalts to the kimberlite pipes is therefore not without justification. Kimberlite pipes are very widely distributed in southern Africa, however, and are of several ages (Allsopp & Barrett, 1975). There is certainly no evidence of a simple linear distribution coincident with the Bouvet plume trace considered here (Fig. 2).

Furthermore, the melilite basalt pipes in the western Cape, and the phonolite lavas and intrusions in the Klinghardt Mountains, dated at around 37 Ma (Kroner, 1973), are further exceptions to the predicted temporal and spatial pattern. It may be, however, that these intrusions belong to a separate province, related to a shorter, more obscure, line of seamounts lying between the Walvis Ridge and the Bouvet–Alphard lineament (Fig. 2). By the hotspot model, the youngest activity along this lineament would occur near Vema Seamount. If these 3 subparallel lineaments owe their origin to hotspots now at Gough, Vema Seamount and Bouvet, respectively, it is clear that the SW coast of Africa passed over the volcanic sources at different times (120, 35 and 60 Ma, respectively).

Assuming the hotspots are stationary features, a pole of rotation for African plate motion relative to the hotspots can be found from the geometry of the lineaments. Such a pole is at 22° 30′ N, 47° 30′ W and small circles of latitude about this pole are superimposed on the volcanic trends in Fig. 2 (for completeness, small circles are also illustrated for Ascension and St Helena hotspots).

Scattered ages along the lineaments can be used to calculate a best-fitting (in the least-squares sense) rate of rotation for Africa about this pole, following the analysis of Duncan & McDougall (1976). Ten-degree intervals of rotation along the small circles are pictured in Fig. 2. Equivalent rotations along the lineaments should predict contemporaneous volcanism. For example, 10° rotation yields approximately 38 Ma old volcanism, 20° rotation yields 55 Ma old volcanism and 30° rotation yields 120 Ma old volcanism.

The geometry and available age information could be compatible with the model of movement of the African plate over fixed sources of volcanism, from Cretaceous time to the present. More geochronology is needed, however, for a rigorous test. Given these data, it would appear that if cratonic igneous activity is related to the passage of a lithospheric plate over a mantle hot spot (or spots) then the manifestation on continental crust is much more varied and complex than on oceanic crust.

Moore (1975) suggested an alternative model for the volcanism in southern Africa. The scarce absolute age determinations together with information on relative ages suggest to Moore (1975) 4 episodes of intense and widespread volcanic activity interspersed with long periods of quiescence since the opening of the proto-Atlantic in the early Cretaceous. The periods of volcanism coincide with breaks in sedimentation (major marine regressions) and could possibly be related to 'epeirogenic upwarping of the continental margins'. The inferred epeirogenic movements may have triggered volcanic activity (or, conversely, the volcanic activity may be the surface manifestation of diapiric upwelling which caused the crustal upwarping).

Moore does not remark on the geographic distribution of ages but it appears that the volcanic centres are concentrated in small areas of different ages and petrologic associations, both seemingly related to their distance from the W coast. The youngest rocks, adjacent to the coast, are olivine melilitites, alone or together with phonolite. Further inland, olivine melilitites are more undersaturated (Brey, 1977) and occur with structures resembling kimberlite pipes and

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with carbonatites (e.g. Gibeon and Bushmanland Province). Diamonds have not yet been found here. The diamond-bearing kimberlites of South Africa occur even further inland and are older (contemporaneous with, and younger than the opening of the Atlantic). Seemingly, there is a relationship between chemistry and distance from the coast, which could perhaps be interpreted in terms of depth of melting. Ages also appear to increase inland from the S and W but geochronology is not sufficient for conclusive modeling. Further work in this area may prove important for the origin of highly undersaturated volcanism on continental crust.

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